

# reSIMATS ENGINEERING

## ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01  
AND EXPERT SYSTEMS

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### COURSE OUTCOME COVERED

**CO3:** *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)

Assessment Tool Weightage: 15%

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**Total Marks: 100**

**Time: 60 Mins**

**No. of Questions: 5**

Annexure A

### Logic Reasoning & Mapping Exercise

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#### **Code of Conduct:**

*I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:** *P Surendra Kumar Reddy*

| Criterion                                           | Weightage | Marks Obtained |
|-----------------------------------------------------|-----------|----------------|
| 1. Problem Analysis and Knowledge Identification    | 15%       |                |
| 2. Logical Representation and Mapping               | 20%       |                |
| 3. Predicates, Variables, Constants and Quantifiers | 15%       |                |
| 4. Logical Connectives and Formula Construction     | 10%       |                |
| 5. Inference and Logical Reasoning                  | 15%       |                |
| 6. Unification and Substitution                     | 10%       |                |
| 7. Resolution and Proof Construction                | 10%       |                |
| 8. Mapping Diagram / Clarity and Presentation       | 5%        |                |
| Total                                               | 100%      |                |

### ANSWER ANY ONE OF THE FOLLOWING

| Q.No | Use Case Scenario                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Smart Healthcare Diagnosis</b> | <p>A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention.</p> <p>Tasks:</p> <ul style="list-style-type: none"> <li>(i) Identify facts, entities, predicates, variables, and constants.</li> <li>(ii) Represent the knowledge using Propositional Logic.</li> <li>(iii) Represent the knowledge using FOL with suitable quantifiers.</li> <li>(iv) Perform unification for two relevant predicates.</li> <li>(v) Apply resolution to prove whether Ravi has a viral infection.</li> <li>(vi) Show the complete inference sequence and final conclusion.</li> </ul>                                                    |
| 2    | <b>Student Academic Advising</b>  | <p>A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support.</p> <p>Tasks:</p> <ul style="list-style-type: none"> <li>(i) Identify the knowledge components.</li> <li>(ii) Convert the statements into Propositional Logic.</li> <li>(iii) Develop FOL representations using predicates and quantifiers.</li> <li>(iv) Identify suitable substitutions and perform unification.</li> <li>(v) Use resolution to prove whether Priya requires academic support.</li> <li>(vi) Explain the reasoning steps leading to the conclusion.</li> </ul> |
| 3    | <b>Intelligent Loan Approval</b>  | <p>A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval.</p> <p>Tasks:</p> <ul style="list-style-type: none"> <li>(i) Identify entities, facts, predicates, and rules.</li> <li>(ii) Map the scenario into Propositional Logic.</li> <li>(iii) Represent the rules using FOL.</li> </ul>                                                                                                                                                             |

|   |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                       | <p>(iv) Demonstrate unification and substitution.</p> <p>(v) Use resolution to prove or disprove LoanApproved(Arun).</p> <p>(vi) Provide the final inference with justification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4 | <b>Vehicle Fault Diagnosis</b>        | <p>An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service.</p> <p>Tasks:</p> <p>(i) Extract the facts and rules from the scenario.</p> <p>(ii) Represent them using Propositional Logic.</p> <p>(iii) Develop suitable FOL predicates and rules.</p> <p>(iv) Perform unification for the vehicle-related predicates.</p> <p>(v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.</p>                                                                              |
| 5 | <b>Cybersecurity Threat Detection</b> | <p>An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach.</p> <p>Tasks:</p> <p>(i) Identify the facts, entities, predicates, and rules.</p> <p>(ii) Map the knowledge into Propositional Logic.</p> <p>(iii) Represent the rules using FOL.</p> <p>(iv) Demonstrate unification and substitution.</p> <p>(v) Apply resolution to determine whether User101 is involved in a possible security breach.</p> <p>(vi) Present the complete reasoning chain and final inference.</p> |

## Q4 – VEHICLE FAULT DIAGNOSIS

### Problem Statement

An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service.

### (i) Extract the Facts and Rules

#### Entities

- Car101 – the vehicle being diagnosed.

#### Facts

1. DoesNotStart(Car101)
2. DimHeadlights(Car101)

#### Predicates

| Predicate            | Meaning                               |
|----------------------|---------------------------------------|
| DoesNotStart(x)      | Vehicle x does not start              |
| DimHeadlights(x)     | Vehicle x has dim headlights          |
| BatteryProblem(x)    | Vehicle x has a battery problem       |
| BatteryInspection(x) | Vehicle x requires battery inspection |
| MarkedForService(x)  | Vehicle x is marked for service       |

#### Constant

- Car101

#### Variable

- x represents any vehicle.

## Rules

### Rule 1:

If a vehicle does not start and has dim headlights, then it has a battery problem.

$$\text{DoesNotStart}(x) \wedge \text{DimHeadlights}(x) \rightarrow \text{BatteryProblem}(x)$$

### Rule 2:

If a vehicle has a battery problem, then it requires battery inspection.

$$\text{BatteryProblem}(x) \rightarrow \text{BatteryInspection}(x)$$

### Rule 3:

If a vehicle requires battery inspection, then it is marked for service.

$$\text{BatteryInspection}(x) \rightarrow \text{MarkedForService}(x)$$

## (ii) Propositional Logic Representation

Let:

- **P** = Car101 does not start
- **Q** = Car101 has dim headlights
- **R** = Car101 has a battery problem
- **S** = Car101 requires battery inspection
- **T** = Car101 is marked for service

## Facts

**P**

**Q**

Therefore:

P= Car101 does not start

Q=Car101 has dim headlights

## Rules

Rule 1:

$$(P \wedge Q) \rightarrow R$$

Rule 2:

$$R \rightarrow S$$

Rule 3:

$$S \rightarrow T$$

## Complete Knowledge Base

$$P, Q, (P \wedge Q) \rightarrow R, R \rightarrow S, S \rightarrow T$$

## Logical Mapping

```
P: Car101 does not start
+
Q: Car101 has dim headlights
↓
R: Car101 has a battery problem
↓
S: Car101 requires battery inspection
↓
T: Car101 is marked for service
```

## (iii) First-Order Logic Representation

First-Order Logic represents the rules generally for all vehicles.

### FOL Facts

*DoesNotStart(Car101)*

*DimHeadlights(Car101)*

## FOL Rules

### Rule 1:

$$\forall x[(\text{DoesNotStart}(x) \wedge \text{DimHeadlights}(x)) \rightarrow \text{BatteryProblem}(x)]$$

### Rule 2:

$$\forall x[\text{BatteryProblem}(x) \rightarrow \text{BatteryInspection}(x)]$$

### Rule 3:

$$\forall x[\text{BatteryInspection}(x) \rightarrow \text{MarkedForService}(x)]$$

Here,  $x$  is a variable representing a vehicle, and **Car101** is a constant. The universal quantifier indicates that the rules apply to all vehicles.

## (iv) Unification and Substitution

Consider the predicates:

$$\text{DoesNotStart}(x)$$

And

$$\text{DoesNotStart}(\text{Car101})$$

For unification:

$$x = \text{Car101}$$

Therefore, the **Most General Unifier (MGU)** is:

$$\{x/\text{Car101}\}$$

Similarly:

$$\text{DimHeadlights}(x)$$

And

*DimHeadlights(Car101)*

give:

$\{x/Car101\}$

### Applying Substitution

Original Rule:

*DoesNotStart(x) ∧ DimHeadlights(x) → BatteryProblem(x)*

Substitute:

$\{x/Car101\}$

Therefore:

*DoesNotStart(Car101) ∧ DimHeadlights(Car101) → BatteryProblem(Car101)*

Since both conditions are true:

*BatteryProblem(Car101)*

## (v) Resolution to Determine Whether Car101 Has a Battery Problem

### Step 1: Convert Rule 1 into Clause Form

Original rule:

*(DoesNotStart(x) ∧ DimHeadlights(x)) → BatteryProblem(x)*

Using:

$A \rightarrow B \equiv \neg A \vee B$

we obtain:

$\neg DoesNotStart(x) \vee \neg DimHeadlights(x) \vee BatteryProblem(x)$



After substituting  $x = \text{Car101}$ :

$$\neg \text{DoesNotStart}(\text{Car101}) \vee \neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$$

## Step 2: Given Facts

$$\text{DoesNotStart}(\text{Car101})$$

$$\text{DimHeadlights}(\text{Car101})$$

## Step 3: Negate the Query

Query:

$$\text{BatteryProblem}(\text{Car101})$$

Negated query:

$$\neg \text{BatteryProblem}(\text{Car101})$$

## Step 4: Resolution

Resolve:

$$\text{DoesNotStart}(\text{Car101})$$

With

$$\neg \text{DoesNotStart}(\text{Car101}) \vee \neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$$

Result:

$$\neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$$

Now resolve with:

$$\text{DimHeadlights}(\text{Car101})$$

Result:

*BatteryProblem(Car101)*

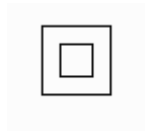
Finally, resolve with the negated query:

$\neg$ *BatteryProblem(Car101)*

Result:

*BatteryProblem(Car101)*

$\neg$ *BatteryProblem(Car101)*



Therefore, a contradiction is obtained, proving that:

*BatteryProblem(Car101)*

## **(vi) Complete Inference Sequence and Final Conclusion**

### **Step 1 – Initial Facts**

*DoesNotStart(Car101)*

*DimHeadlights(Car101)*

### **Step 2 – Apply Rule 1**

Since Car101 does not start and has dim headlights:

*BatteryProblem(Car101)*

### **Step 3 – Apply Rule 2**

Since Car101 has a battery problem:

*BatteryInspection(Car101)*

#### Step 4 – Apply Rule 3

Since Car101 requires battery inspection:

*MarkedForService(Car101)*

#### Complete Inference Diagram

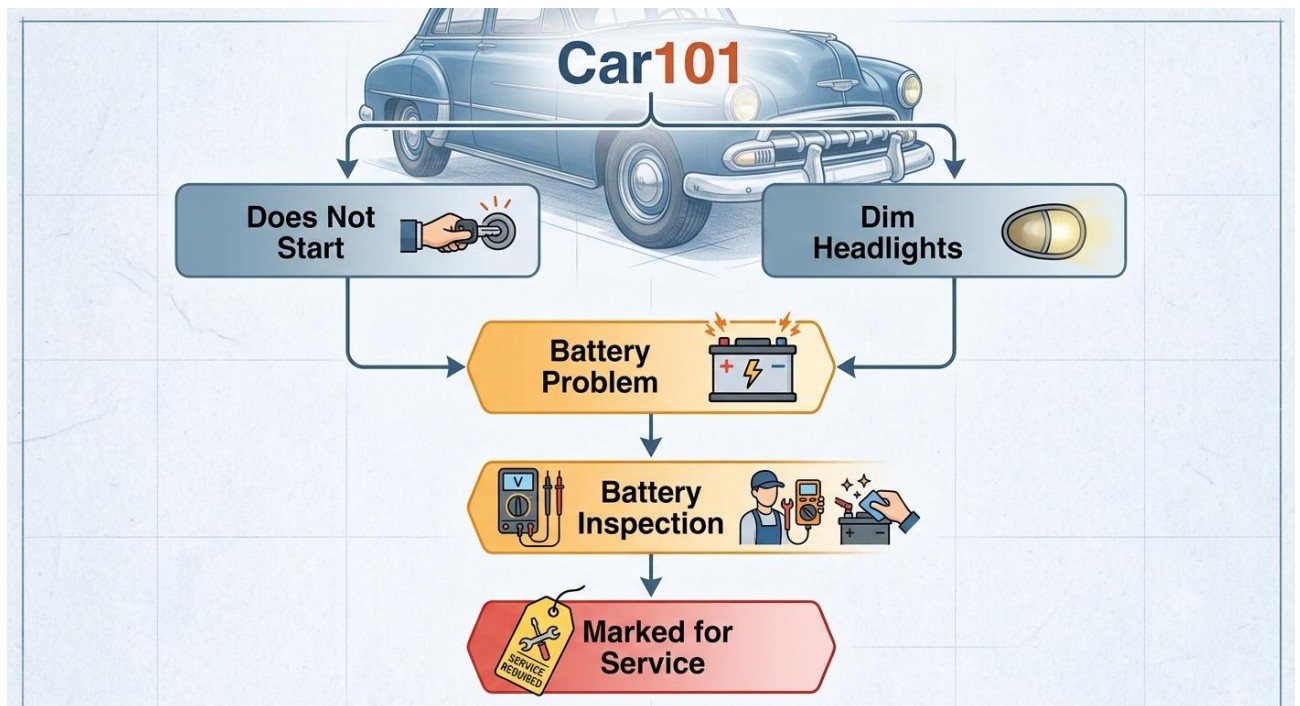


Figure 1: Car Battery Troubleshooting Flowchart

## Final Conclusion

The expert system concludes that **Car101 has a battery problem** because it does not start and has dim headlights.

Therefore:

*BatteryProblem(Car101)*

The battery problem leads to:

*BatteryInspection(Car101)*

and consequently:

*MarkedForService(Car101)*

## Final Inference

- Car101 does not start + dim headlights → Battery Problem → Battery Inspection → Marked for Service.

## EVALUATION RUBRICS:

| Criterion                                        | Weightage (%) | Excellent                                                                                                                    | Good                                                                 | Average                                                                  | Poor                                                            |
|--------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------|
| Problem Understanding & Knowledge Identification | 15%           | 13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario. | 10–12% – Identifies most relevant knowledge with minor omissions.    | 7–9% – Identifies basic knowledge but misses several important elements. | 0–6% – Poor or incorrect identification of the given knowledge. |
| Logical Representation & Mapping                 | 20%           | 17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.         | 13–16% – Provides mostly correct logical mappings with minor errors. | 9–12% – Provides partial mappings with several logical errors.           | 0–8% – Logical mapping is largely incorrect or missing.         |
| Predicates, Variables, Constants & Quantifiers   | 15%           | 13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.           | 10–12% – Uses the concepts correctly with minor errors.              | 7–9% – Demonstrates partial understanding with some incorrect usage.     | 0–6% – Concepts are incorrectly applied or missing.             |
| Logical Connectives & Formula Construction       | 10%           | 9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to                                                      | 7–8% – Mostly correct use of connectives with minor errors.          | 5–6% – Basic use of connectives but several errors occur.                | 0–4% – Incorrect or incomplete formula construction.            |

|                                         |     |                                                                                                                |                                                                     |                                                                    |                                                               |
|-----------------------------------------|-----|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|
|                                         |     | construct valid logical formulas.                                                                              |                                                                     |                                                                    |                                                               |
| Inference & Logical Reasoning           | 15% | 13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.      | 10–12% – Reasoning is mostly correct with minor gaps.               | 7–9% – Provides partial reasoning with some incorrect conclusions. | 0–6% – Reasoning is incorrect, unsupported, or missing.       |
| Unification & Substitution              | 10% | 9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.         | 7–8% – Correctly performs most unification steps with minor errors. | 5–6% – Demonstrates basic understanding but has several errors.    | 0–4% – Unable to perform unification correctly.               |
| Resolution / Proof Construction         | 10% | 9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.                 | 7–8% – Resolution is mostly correct with minor errors.              | 5–6% – Shows partial understanding but proof contains gaps.        | 0–4% – Resolution is incorrect or not demonstrated.           |
| Clarity, Mapping Diagram & Presentation | 5%  | 5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation. | 4% – Presentation is clear with minor formatting or diagram issues. | 3% – Basic presentation with limited clarity.                      | 0–2% – Poorly organized, unclear, or incomplete presentation. |